

Harshvardhan Singh

Data Scientist

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Summary

Machine Learning & AI Engineer with 4+ years of experience in developing scalable AI solutions across machine learning, deep learning, NLP, and computer vision. Proficient in Python, TensorFlow, PyTorch, Scikit-Learn, and cloud platforms like AWS SageMaker, and Azure ML. Skilled in MLOps, CI/CD, Kubernetes, and Docker for seamless model deployment. Expertise in big data processing (Spark, Hadoop, Databricks), real-time analytics (Kafka, Airflow, Snowflake), and database management (SQL, NoSQL, GraphDBs). Passionate about AI ethics, explainability, and model optimization, driving impactful AI-driven decision-making and business transformation.

Skills

Languages:	Python, R programming, C/C++, SQL
Libraries & Model:	TensorFlow, PyTorch, Keras, Scikit-learn, Pandas, NumPy, Seaborn, Matplotlib, Snorkel, Hugging face, Lang Chain, Lang Graph, NLTK, SpaCy, OpenCV, Transformers (BERT, GPT, RoBERT)
Databases:	MySQL, PostgreSQL, Oracle, MongoDB
Cloud Services:	AWS, Microsoft Azure, GCP
Artificial Intelligence:	Supervised & Unsupervised Learning (Regression, Classification), Reinforcement Learning, Deep learning Architectures (Neural networks, Transfer Learning), Reinforcement Learning, Computer Vision, Natural Language Processing (Semantic Analysis, Text segmentation), ML Ops (ML Flow), LLM's (GPT-4, Claude, Time Series Analysis
Data Engineering:	PySpark, Apache Airflow, Data Bricks, ETL/ELT Pipelines, Data Modeling, Hadoop, Teradata, Big Query, Snowflake
Tools:	DataBricks, Tableau, Power BI, Retool, Spark, Docker, Google Colab, GitHub, Any labeling
Others:	MS Office (Word, Excel, PowerPoint), SDLC, Agile Development, Jira, CI/CD Pipelines

Experience

Data Scientist - Staff Member

| May 2024 – Present

University of Arizona, Tucson, USA

- Designed and implemented a custom ETL pipeline by AWS to extract 27M research abstracts from OpenAlex, leveraging Athena for efficient SQL querying, S3 for scalable storage, Python & R scripts for data transformation and preprocessing and executing batch jobs via SLURM on an HPC terminal by 'subprocess' for automated shell command execution cutting runtime by over 80%
- Performed NLP text processing tasks including key phrase extraction using SpaCy and RAKE, to generate n-grams (up to trigrams) from raw research abstracts for input into topic modeling algorithms
- Trained multiple topic models (statistical models in Natural Language Processing), and addressed key challenges such as avoiding the dummy variable trap in the regression models within STMs, etc.
- Leveraged Python & R to build, dissect and interpret our STM, Labeled LDA and DTM (with transformer based BERTopic) models to study the influence of countries on research topics, focusing on themes such as the Matthew Effects (rich-getting-richer) and the distribution dynamics of topical "stock portfolios", using the Herfindahl-Hirschman Index (HHI) and Entropy to quantify concentration and topical dominance
- Constructed a bipartite country–topic influence matrix by computing cosine similarity-based influence scores derived from topic distributions of Dynamic Topic Models (DTMs) and Labeled LDA, while optimizing the alignment of topic term vectors with country-level distributions to ensure accurate semantic influence measurement
- Constructed a country–country cross similarity matrix using pairwise cosine similarity on topic influence vectors; analyzed temporal patterns by grouping countries into core and peripheral clusters using HHI and computed intra- and inter-group similarities to study global research convergence trends
- Navigated through iterative visualization phases to elucidate trends like countries punching above/below their weight, delivered comprehensive insights through dual thematic plots
- Facilitated new research team members to integrate with the project by explaining the research objectives, outlining finer details and providing technical guidance on the tools and data protocols used, while also familiarizing myself with their individual research contributions within the broader project scope
- Achievement: Successfully upscaled our study by implementing the data pipeline that enabled the analysis of multiple research fields, significantly expanding the scope of our research**

Data Scientist - Graduate Research Assistant

| May 2023 – May 2024

The Global Knowledge Lab and Observatory, University of Arizona, USA

- Carried out research tasks on HPC, while maintaining high core hour and job efficiency of over 80% and managed repositories on the HPC using Shell Scripting
- Conducted diagnostics including Held-Out Likelihood, Residuals, Semantic Coherence and Lower Bound to estimate the optimal value for number of topics (K) to be generated, ensuring high interpretability of the topics
- Coordinated with the Supervisor to take several critical research decisions from data preprocessing phase to model training phase
- Built a classifier using a weak supervision framework with 'Snorkel' to classify unlabelled data with noisy set of labels
- Extracted big datasets via AWS comprising of CSV files (500,000+ rows) and JSONL files consisting of titles, abstracts, full-text, n-grams and metadata, and preprocessed the data, improving the quality of topics generated and reducing iterations required for model convergence from 250+ to 43 for the STM model with encoded covariates
- Manually labelled 500+ rows of dataset to build CNN classifier which worked as a downstream classifier for the 'Snorkel' model

- Created edge lists and performed Network Analysis with graphs for citations, co-authorships and collaborative networks while utilizing the 'networkx' library to evaluate key graph metrics such as centrality, clustering coefficients, connectivity and degree distribution while expanding analysis by implementing cutting-edge graph embedding algorithms like Node2Vec & Metapath2Vec to extract high-dimensional data insights
- Preprocessed citation data and calculated Citation Disruption Index by complex math formulas while handling exceptions, and integrated it with the other dataframes to create topic level, covariate level and document level summarized dataframes for comprehensive analysis
- **Achievement: Given the promising results, the project has received a Department of Defence (DoD) research grant, and is now expanding in scope to perform this analysis on hundreds of fields beyond just Astronomy/Astrophysics**

Data Scientist

| Jan 2021 – Jul 2022

HCL Tech, India

- Performed exploratory data analysis (EDA) to uncover trends and derive actionable insights that supported strategic decisions in risk analytics and operational planning.
- Collaborated with cross-functional teams to translate complex business problems into data science use cases, driving measurable improvements in revenue and efficiency.
- Designed and deployed customized credit scoring models tailored to specific industry verticals, enhancing clients' risk assessment and forecasting accuracy.
- Utilized Natural Language Processing (NLP) techniques to extract key themes and sentiment from customer service transcripts, contributing to a 20% uplift in customer satisfaction scores.
- Applied clustering and other unsupervised learning methods to segment credit card customers dynamically, leading to increased precision in targeted marketing campaigns.
- Created end-to-end ML pipelines, leveraging AWS Glue, Lambda, and S3, to collect, clean, and preprocess over 10 million records from Electronic Health Records (EHRs) and hospital management systems, improving data accuracy by 25%.
- Built a hybrid forecasting model combining ARIMA, LSTMs, and XGBoost, increasing demand forecasting accuracy by 35%, and scaled model training using Apache Spark on AWS EMR for large-scale data processing.
- Implemented a real-time inventory recommendation system using Kafka and AWS Kinesis, reducing latency by 45%, and enhanced model explainability with SHAP, TensorBoard, and AWS CloudWatch for reliable decision-making.
- Conducted rigorous model validation, feature selection, and hyperparameter tuning to ensure model robustness, compliance, and consistent performance.
- Built interactive dashboards and performance reports using Tableau, delivering key business intelligence insights to leadership and enhancing data-driven decision-making.

Education

Master of Science in Data Science - University of Arizona, AZ

| May 2024

Bachelors of Technology in Computer Science and Engineering - SRM Institute of Science and Technology, India

| May 2022

Certifications

Generative AI for Data Scientists Specialization (IBM)

Google Advance Data Analytics Certification

Advance Statistics for Data Science Specialization (Johns Hopkins Uni.)

SQL Advanced Cert. (HackerRank)

Project Exposition 2022 Certification (SRMIST)

Research Day Conference (SRMIST)

D2C Hackathon Festival – 22nd Rank (6701 participants)

Projects

Analyzing Country Influence in Research Fields Using DTM (with BERTopic) and Labeled LDA

| Sep 2024 - Present

Project Portfolio - <https://www.harsh-singh.net/about-me/project-two>

Foxes vs Hedgehogs: Classifying Countries as Generalists or Specialists Based on Entropy Scores

| May – Sep 2024

Project Portfolio - <https://www.harsh-singh.net/about-me/project-three>

Mapping Research Dynamics: Analyzing Topics & Country Contributions Using Structural Topic Models

| 2023 - 2024

Project Portfolio - <https://www.harsh-singh.net/about-me/project-four>

Automating Data Source Labeling using Snorkel Framework for Weak Supervision

| Oct – Nov 2023

Project Portfolio - <https://www.harsh-singh.net/about-me/project-5>

Graph Networks in Global Research: Citation Connectivity, Clustering, and Centrality Analysis

| May – Nov 2023

Project Portfolio - <https://www.harsh-singh.net/about-me/project-6>

Skin Cancer Progression Prediction

| Oct 2022 – Apr 2023

Project Portfolio - <https://github.com/ISTA421INFO521/Skin-Cancer-Progression>

Classifying Astronomical Transients

| Oct – Dec 2022

Project Portfolio - <https://github.com/astrochialinko/INFO523-Final-Project/tree/Harsh>

Conferences/Publications

“The Centralization of Global Research Focus Is Associated with Declining Disruption in Fields” Accepted for presentation at the 4th International Conference on the Science of Science and Innovation (ICSSI)

| Jun 16 - 18, 2025

Extended Abstract - <https://drive.google.com/file/d/1LDieLiC4PdAWtAJ4Bkngh2rdTh71W0DI/view?usp=sharing>

“Centralization of Research Influence and Declining Novelty: Text-Based Approaches to Global Scientific Hegemony” Accepted for presentation at the 120th ASA Annual Meeting

| Aug 8 – 12, 2025

Extended Abstract - <https://drive.google.com/file/d/1Lw58cxqNyl5UwXhe6AtLsNOagPRQrTpG/view?usp=sharing>